

Networking Protocols

Lesson 41 — Protocols, OSI Layers and Addresses

JKSSB Computer Awareness

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Why Protocols Exist

- ▶ Computers on a network follow agreed rules so that data sent by one machine can be understood by another.
- ▶ A **protocol** is such a set of rules for communication: what messages look like and in which order they are exchanged.
- ▶ Different jobs need different protocols: moving files, opening web pages, sending email, and finding addresses each use their own protocol.
- ▶ Every protocol has an **exact English name**; the exam tests the name and its role together.

The one idea

One protocol, one job: match each name (TCP, DNS, HTTP) to the single task it performs.

TCP vs UDP: The Transport Pair

Point	TCP	UDP
Full form	Transmission Control Protocol	User Datagram Protocol
Delivery	Reliable, ordered, error-checked	Fast, best-effort, no guarantee
Connection	Connection-oriented	Connectionless
Typical use	Web, email, file transfer	Streaming, DNS queries, voice calls

TCP guarantees order, not speed; UDP gives up guarantees for speed.

TCP Gives Order, Not Timing

- ▶ **TCP** numbers every segment and retransmits lost ones, so bytes arrive complete and **in order**.
- ▶ TCP does **not** promise when data arrives — only that it arrives correctly ordered. (PYQ-18)
- ▶ **UDP** sends datagrams with no connection, no ordering, and no retransmission.
- ▶ Use TCP when correctness matters (a file, a page, a message); use UDP when speed matters more (live audio, video, DNS lookups).

Trap alert

TCP guarantees **ordering**, never timing. “TCP delivers fast” is a wrong answer.

IP: Every Machine Needs an Address

- ▶ The **Internet Protocol (IP)** carries addressed packets from the sender to the receiver across networks.
- ▶ Each device gets an **IP address**; routers read it to forward the packet hop by hop.
- ▶ IP by itself is **unreliable**: delivery is best-effort, and TCP sitting above it adds the reliability.
- ▶ Two versions exist: **IPv4** (32-bit) and **IPv6** (128-bit).

Keep the pair straight

IP finds the address; TCP makes the delivery reliable.

IPv4 vs IPv6

Point	IPv4	IPv6
Length	32-bit	128-bit (TRM-10)
Written as	Four decimal octets	Eight hexadecimal groups
Example	192.168.1.10	2001:0db8::1
Header	Variable, 20–60 bytes	Fixed at 40 bytes

Trap alert

IPv6 is **128-bit**, never 256-bit. (TRAP-18)

ICMP: Control, Not Data

- ▶ **ICMP** stands for Internet Control Message Protocol.
- ▶ It carries **control and diagnostic** messages about the network itself: errors and reachability reports.
- ▶ Its tools are **ping** (is the host reachable?) and **traceroute** (which path do packets take?).
- ▶ ICMP transfers **no application data** and uses **no port numbers**. (TRAP-19)

Office desktop example

When a page will not load, a ping tests whether the network path works; the ping itself carries no page content.

DNS: Names to Numbers

- ▶ **DNS** stands for Domain Name System.
- ▶ It translates a human name such as `jkssb.nic.in` into the **IP address** the network needs.
- ▶ Without DNS, every user would have to memorise numeric addresses.
- ▶ DNS runs mainly over **UDP** on port **53**, falling back to TCP for large transfers.

One-line recall

DNS = the phone book of the Internet: names in, numbers out.

DHCP: Automatic Addresses

- ▶ **DHCP** stands for Dynamic Host Configuration Protocol.
- ▶ It **automatically assigns** an IP address, subnet mask, gateway, and DNS server to each device that joins the network.
- ▶ Without DHCP, every address would have to be typed in by hand.
- ▶ A home router and an office network both run a DHCP server; the joining laptop or phone is the DHCP client.

DNS vs DHCP

DNS resolves names to addresses; DHCP hands out addresses to newcomers. Do not swap them.

HTTP vs HTTPS: The Web Pair

Point	HTTP	HTTPS
Full form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Port	80	443
Security	Plain text	Encrypted with SSL/TLS
Shows as	http://	https:// + padlock

HTTPS is HTTP plus an SSL/TLS security layer; the S stands for Secure.

FTP, Telnet, SMTP: Three More Roles

- ▶ **FTP** (File Transfer Protocol, port 21) moves files between computers; it is not email and not web browsing.
- ▶ **Telnet** (port 23) opens a plain-text remote login session; it is replaced today by the encrypted SSH.
- ▶ **SMTP** (Simple Mail Transfer Protocol, port 25) **sends** outgoing mail to the server.
- ▶ Receiving mail is a different job: **POP3** (port 110) downloads it, **IMAP** (port 143) keeps it synced on the server.

Send vs receive

SMTP sends; POP3 and IMAP receive. Do not swap them.

The OSI Model: Seven Layers

No.	Layer	One-line job
7	Application	Services the user sees (HTTP, FTP, SMTP, DNS)
6	Presentation	Translation, encryption, compression
5	Session	Opens and manages sessions between machines
4	Transport	Reliable or fast delivery (TCP, UDP)
3	Network	Addressing and routing (IP, ICMP)
2	Data Link	Frames on one link (switch, MAC address)
1	Physical	Bits on the wire or air

Read top-down for services, bottom-up for the journey of one bit.

Where Each Protocol Lives

- ▶ **Application** layer: HTTP, HTTPS, FTP, Telnet, SMTP, POP3, IMAP, DNS, DHCP.
- ▶ **Transport** layer: TCP and UDP — the only two names to place here at this exam depth.
- ▶ **Network** layer: IP (IPv4, IPv6) and ICMP.
- ▶ Data moves **down** the layers on sending and **up** the layers on receiving; each layer adds its own header.

Worked mapping

Opening a web page uses HTTP at layer 7, TCP at layer 4, and IP at layer 3 — one protocol per layer, working together.

Port Awareness: Numbers Name the Service

- ▶ A **port number** tells the receiving machine which service a packet is for; the IP address names the machine, the port names the service.
- ▶ Ports 0–1023 are the **well-known ports** every candidate must recognise.
- ▶ FTP 21, Telnet 23, SMTP 25, DNS 53, HTTP 80, POP3 110, IMAP 143, HTTPS 443.
- ▶ ICMP has **no ports at all**; DHCP uses UDP ports 67 and 68.

Exam drill

Cover the service names, recall the numbers: 21–23–25–53–80–443.

Near-Neighbour Contrasts

- ▶ TCP is **reliable and ordered**; UDP is **fast and best-effort**.
- ▶ HTTP is **plain**; HTTPS adds **SSL/TLS encryption**.
- ▶ FTP moves **files**; SMTP sends **mail**; Telnet opens a **remote session**.
- ▶ DNS **resolves names**; DHCP **assigns addresses**; ICMP reports **errors**.
- ▶ IPv4 is **32-bit dotted decimal**; IPv6 is **128-bit hexadecimal** with a **40-byte** header.

Trap Alert: Protocol Facts to Keep Straight

- ▶ IPv6 = **128-bit**, never 256-bit; header fixed at 40 bytes. (TRAP-18, TRM-10)
- ▶ ICMP is **control/diagnostic** only: no data, no ports. (TRAP-19)
- ▶ TCP promises **order**, not timing; UDP promises neither. (PYQ-18)
- ▶ SMTP **sends** mail; POP3/IMAP **receive** it.
- ▶ DNS resolves names; DHCP assigns addresses; HTTP is port 80 and HTTPS is port 443.

Lesson 41: Recall

- ▶ TCP is reliable, ordered, and connection-oriented; UDP is fast, connectionless, and best-effort.
- ▶ IPv4 is 32-bit; IPv6 is 128-bit with a fixed 40-byte header.
- ▶ ICMP carries control and diagnostic messages (ping, traceroute); it transfers no data and uses no ports.
- ▶ DNS translates names to IP addresses; DHCP automatically assigns addresses to joining devices.
- ▶ HTTP (port 80) fetches web pages; HTTPS (port 443) adds SSL/TLS; FTP (21) moves files; Telnet (23) gives remote login; SMTP (25) sends mail.
- ▶ The OSI layers run Physical, Data Link, Network, Transport, Session, Presentation, Application; ports name the service on a machine.